supply, expressed as amount of food per bird, is poor. Whether or not we will observe the predicted positive correlation between return rate and amount of food per bird depends on the time scale over which the local population becomes adjusted to a lowered or increased food supply. If the population becomes adjusted within a single winter, we will not observe a significant correlation between return rate and food supply. In that case, the return rate will not depend on the mismatch between the number of birds and the food supply in the current year, but on the difference between the food stock in the current year and the next year. If, on the other hand, adjustment takes many years, as seems to be the case for the oystercatchers in the Dutch Wadden Sea, we may not observe a correlation either. The slow response of the oystercatchers in the Dutch Wadden Sea may be explained by the fact that they comprise more than half of the continental European population (Goss-Custard et al., 1996a). Thus, changes in the Wadden Sea population will be largely determined by natality and mortality, instead of immigration and emigration. Both natality and mortality are low in the long-lived oystercatcher, so adjustment will be slow, except when extreme events cause high mortality. Thus, the absence of a significant correlation cannot be taken as evidence that the population does not respond to changes in the size of the food stock. Uncertainty in the estimates of the population size, related to imputing, poses an additional problem. The estimated trend in the total number of wintering oystercatchers is quite robust to the precise details of imputing. However, return rate depends on the difference in numbers between two consecutive years. Since oystercatchers population numbers generally change slowly over the years, this difference is generally small. As a result, small differences in the estimated total number of oystercatchers, which do not matter for the overall trend, can have a big effect on the estimated return rate. According to recent insights it is better to impute missing data on the basis of individual counting sites and not first clustering these counting sites into larger units (Soldaat et al., 2004). Preliminary calculations indicate that return rate is quite sensitive to the method of imputing and that the new method adopted by SOVON leads to poorer correlations for the Dutch Wadden Sea than the correlation depicted in Figure 7411. Given the uncertainties surrounding the estimate of return rate for the Dutch Wadden Sea, it was necessary to invest most effort in developing a model describing how oystercatchers deplete a given food stock and whether they are stressed for food in the process. A short description of the model, including the most important parameters and assumptions is provided in section 1.5.7. For a detailed account we refer to (Rappoldt et al., 2003d). The amount of food stress calculated with the model increases with a decreasing food supply. In addition, and as expected, the return rate 11 The problems with estimating return rate in the Dutch Wadden Sea do not apply to the Oosterschelde. First, the entire Oosterschelde is counted every month, so the extent of imputing needed is small compared to the Wadden Sea. Second, the Oosterschelde is much smaller, so changes in numbers between years may depend more on immigration and emigration than mortality and reproduction. Thus, adjustment to a changed food supply may occur on a shorter time scale compared to the Wadden Sea. This may explain why the relationship between return rate on the one hand and food supply and food stress on the other hand is much tighter for the Oosterschelde than for the Wadden Sea. 